FANTASY DRAFT AID

CS 465

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INTRODUCTION

- Fantasy Premier League is an online game where millions of participants draft real-life Premier League players into fantasy squads to score points based on weekly performances.
  - Draft is a game mode of premier league fantasy. A league of typically 8-12 managers compete head-to-head using unique teams to try and end up at the top of the league table once the final gameweek has finished.
- In a competitive league, hours and hours of research must be done to facilitate draft success. One mistake or unlucky circumstance can derail a years' worth of work.
- The goal of this project was to help managers make informed decisions while not needing to contemplate or debate the future implications a trade may have

KEY RELATED WORK

- Implementing AI to make predictions in fantasy sports, as well as their real-life counterparts, is a well-researched problem, many of which explore using Random Forest Regressors to make predictions on player performance
  - Player Recommendation System for FPL
    - Demonstrated the use of classification and regression models to recommend FPL players based on position, form, and price [1].
    - Inspired the two-model architecture: short-term vs. long-term performance predictions.
  - Random Forest-Based Fantasy Football Team Selection
    - Applied Random Forest Regression to weekly team selection, validating its ability to model FPL performance [2].
    - Supported our choice to use Random Forest as the predictive core for both scoring models.
  - Using ML Models to Predict Points in FPL
    - Compared various ML models (Random Forest, Gradient Boosting, etc.) and analyzed feature importance [3].
    - Encouraged our emphasis on feature selection and interpretation for transparency in predictions.
- While most of these methods have focused on classic FPL and traditional fantasy sports formats, this project adapts and extends them specifically for the under-served draft format.

APPROACH

- The purpose of this project is to try and predict how well each player will perform in the near future, then extrapolating the data to give an accurate estimation of how well the player will do over the course of the season.
- Short-Term Model
  - Predicts next game week's score.
- Long-Term Model
  - Predicts rest of season (ROS) performance
  - Helps to assess the total value of players in trade negotiations.
- These models are used to:
  - Rank free agents
  - Assess trade targets and fillers
  - Score team strength over time

APPROACH

- Also wanted the model to be capable of recommending realistic, balanced trade offers between managers.
- Searches for high-value players owned by other teams
- Proposes a 3-1 trade package, i.e. one manager trades 3 mediocre players for 1 premium, and two lower tier players fill out the numbers.
- Ensures total net gain is reasonable and realistic, within 10 projected points
- Filters trades by
    - Position compatibility
    - Manager ownership
    - ROS projections
  - to provide explainable and actionable suggestions
- A Deep Q-Network was originally the choice to implement this feature, but there were unfortunately some issues

EXPERIMENTS & TRAINING

- Data Sources
  - FPL official API – player stats, fixture difficulty, ICT index, etc.
    - FPL API is also used to access information on managers and teams within the league
    - A dictionary is kept of managers and their rosters,
  - xG and xGA from Understat to add to feature calculation
  - Historical FPL data from publicly available on github

EXPERIMENTS & TRAINING

- Short-Term Model
  - Random Forest Regressor from scikit-learn
  - Features:

```
features = ['form','ict_index','influence','creativity','threat',
            'minutes','goals_scored','assists','clean_sheets',
            'yellow_cards','red_cards','team_form','fixture_difficulty']
```

  - Training Data: gameweek-by-gameweek format, each row = a player-GW
  - Target: next game-week's points
- Long-Term Model
  - Random Forest Regressor from scikit-learn
  - Features:

```
features = ['goals_scored','assists','ict_index','influence','creativity',
            'minutes','team_form','fixture_difficulty']
```

  - Target: projected rest of season (ROS) points.
- Feature Engineering and target variable selection allow for the models to differentiate in purpose and outcome

RESULTS

- The Short-Term Model

```
=== Training short-term model ===
[short] CV MAE: 0.954 ± 0.147
[short] Hold-out MAE: 1.151, RMSE: 2.399, R²: 0.246
```

- The Long-Term Model

```
=== Training long-term model ===
[long] CV MAE: 1.961 ± 0.318
[long] Hold-out MAE: 2.299, RMSE: 4.072, R²: 0.990
```

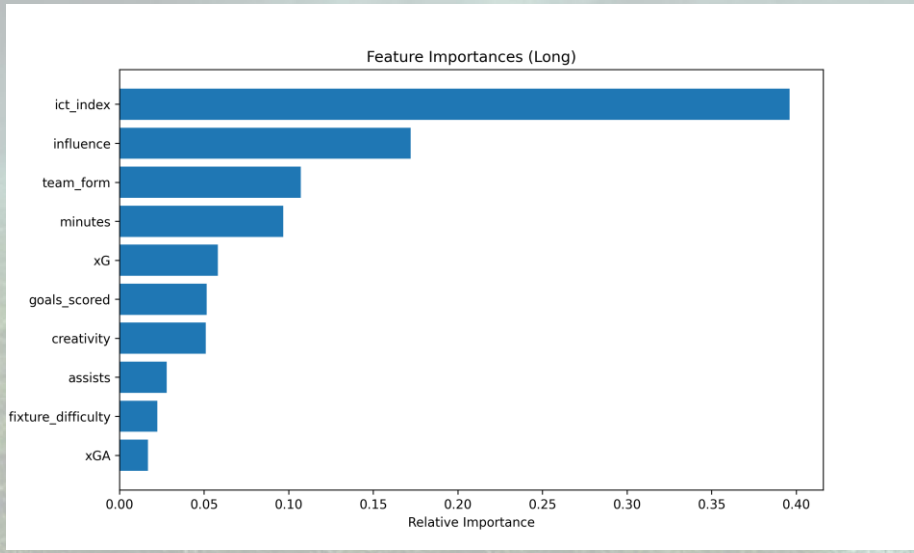
- K Fold
  - Splits dataset into k-sized folds, randomly shuffling before partitioning
  - Runs k times. In each run, one fold is held out for validation while other k-1 folds are used for training using the Random Forest Regressor
  - Used for short term as it allows for generalization across all gameweeks
- Time Series Split
  - Splits the folds chronologically, fold one trains on the earliest points and validates thereafter, fold two extends the training window and so on.
  - As we're trying to predict future performance over a prolonged period, the Long-Term model can validate the predictions of each training slice immediately on known data to better it's estimation.

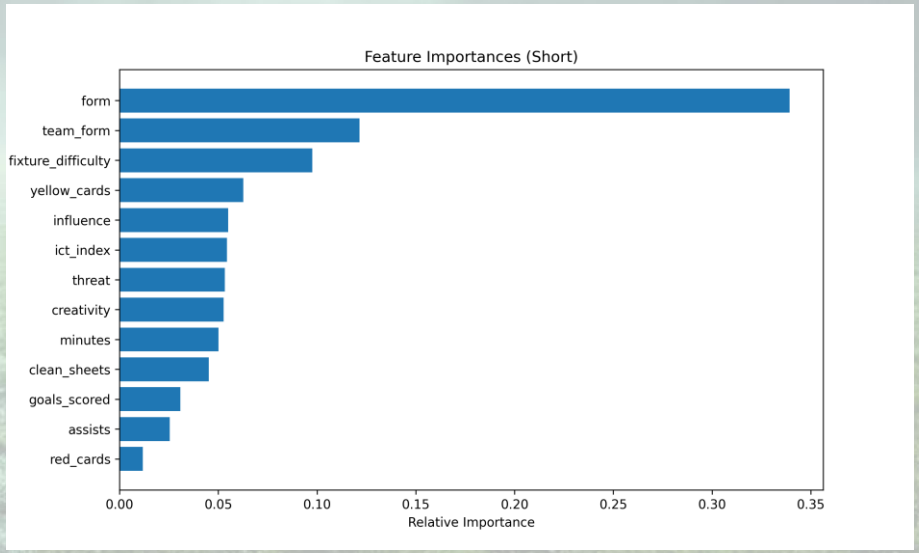
RESULTS

- Metrics
  - CV MAE (Cross Validation Mean Absolute Error)
    - The average absolute error computed across all cross-validation folds, where each fold is held out in turn
      - For the short-term model, the estimation is within ~ 1 point
      - For the long-term model, the estimation seems to be outstanding, projecting within ~ 2 points,
        - Variance is limited with the season ending soon
  - MAE
    - The average of absolute differences between predicted and actual points
      - Short-Term is 1.15 points
      - Long-Term is ~ 2.5 points as the players current points are added in to make the forecasting projection
  - RMSE (Root Mean Squared Error)
    - The square root of the mean squared errors, penalizes large misses
      - Short-Term is ~2.4 points,
      - Long-Term is ~ 4 points.
  - R^2 (Coefficient of Determination)
    - Fraction of variance in the true scores that the model explores
      - We’re able to explain ~ 25% of the variance in the short-term model, inherent randomness of the problem limits the projection for a single gw,
      - Long-Term again benefits from the player’s current points being applied to the forecast, explaining ~99% of the variance.

RESULTS

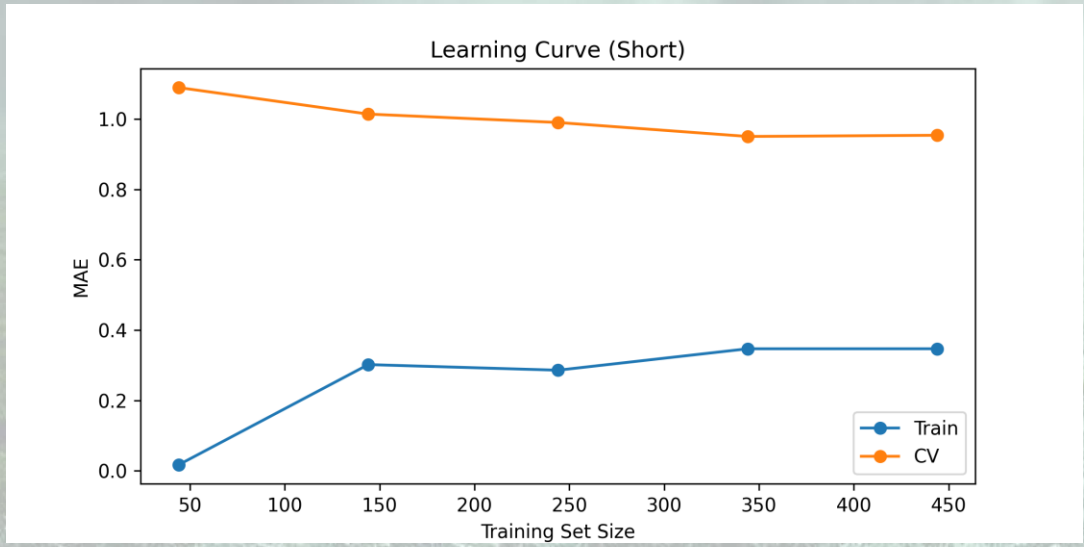
- Feature Importance
  - Displays which features each model resonated towards to best root out variance in its estimations

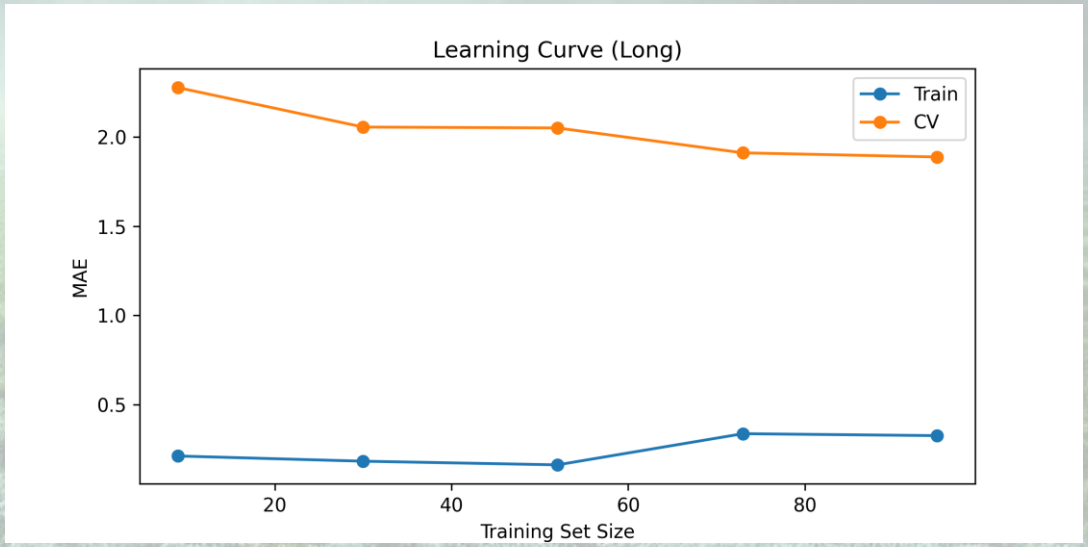




RESULTS

- Learning Curve
  - The MAE for each model after player-gw rows used for training
    - “Train” is the MAE on the training folds
    - CV is the MAE on the held-out, validation, folds





RESULTS

My FPL Team Dashboard

My Team | Free Agents | Trade Suggestions | Trade Analyzer

Current Roster and Projections

	Name	Team	Pos	Points So Far	↓ Predicted GW35	Predicted ROS	Projected Total
236	Sánchez	Chelsea	GK	110	7.48	20.6312	130.6312
712	Schär	Newcastle	DEF	104	6.95	17.4	121.4
582	Gakpo	Liverpool	FWD	107	4.9	18.0753	125.0753
818	Onuachu	Southampton	FWD	49	2.26	6.0471	55.0471
140	Huijsen	Bournemouth	DEF	91	2.02	8.8606	99.8606
129	O.Dango	Bournemouth	MID	110	1.89	11.0353	121.0353
31	Trossard	Arsenal	MID	136	1.78	8.4453	144.4453
323	Ndiaye	Everton	FWD	96	0.88	7.8329	103.8329
898	Pedro Porro	Spurs	DEF	99	0.62	6.7041	105.7041
270	Henderson	Crystal Palace	GK	123	0.56	7.9053	130.9053

Predicted GW35 Team Score

30.1 pts

My FPL Team Dashboard

My Team | Free Agents | Trade Suggestions | Trade Analyzer

Top Free Agents for GW35

Select Position Group

DEF

	Player	Team	GW35 Predicted	Points So Far
406	Sessegnon	Fulham	9.2900	54
967	Toti	Wolves	4.3600	75
971	Agbadou	Wolves	2.6700	35
816	Manning	Southampton	1.9900	32
388	Tete	Fulham	1.8700	46
376	Bassey	Fulham	1.8300	72
338	O'Brien	Everton	1.7900	52
373	Andersen	Fulham	1.7600	47
685	Yoro	Man Utd	1.7400	21
211	Gusto	Chelsea	1.7000	56

Waiver Comparison

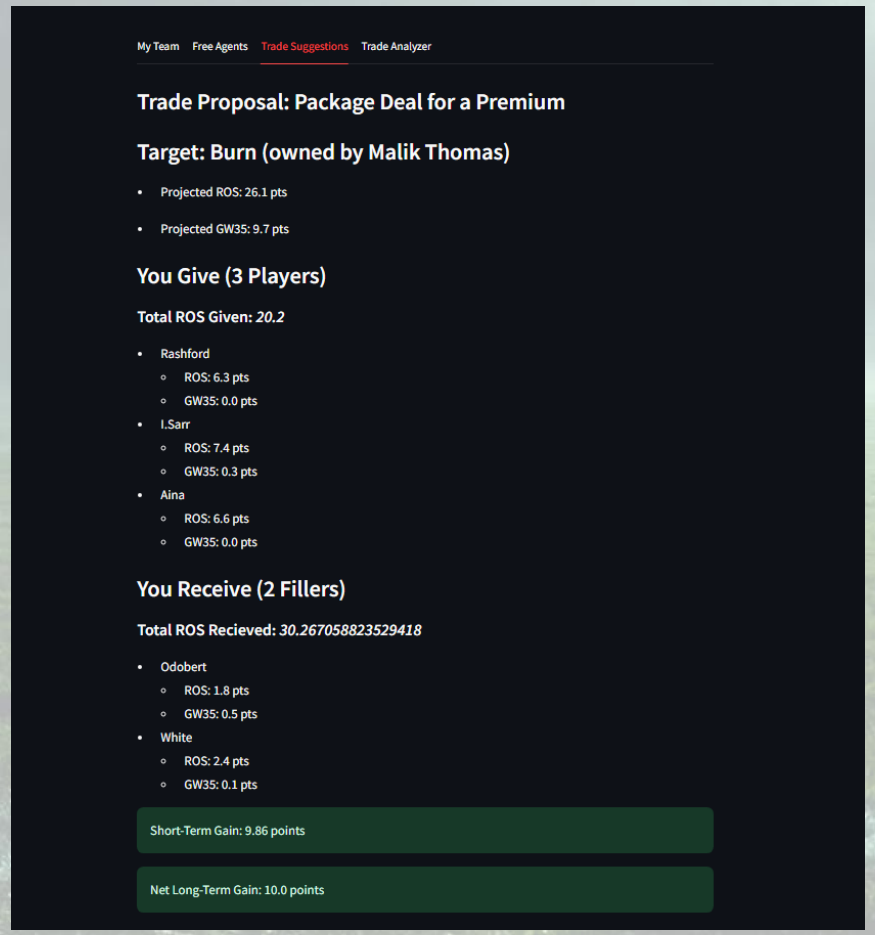
Select a Player From Your Team

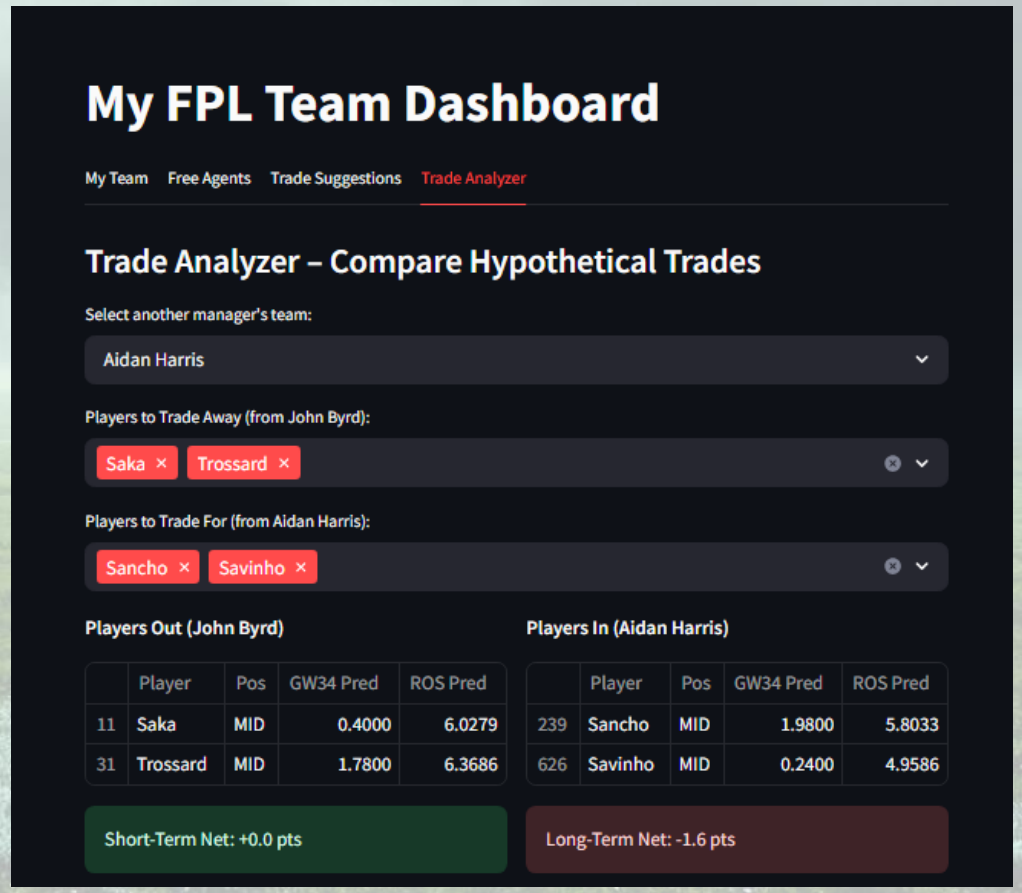
Henderson

Compare With Free Agent (GK)

Hermansen

RESULTS





	Player	Pos	GW34 Pred	ROS Pred
11	Saka	MID	0.4000	6.0279
31	Trossard	MID	1.7800	6.3686

	Player	Pos	GW34 Pred	ROS Pred
239	Sancho	MID	1.9800	5.8033
626	Savinho	MID	0.2400	4.9586

CONCLUSION

- Using Random Forest Regression, the system
  - **Predicts short-term player performance**
  - **Estimates long-term value** based on projected season totals
  - **Identifies optimal waiver signings** to patch positional gaps
  - **Suggests realistic trades** that benefit the manager
  - **Evaluates trades** based on projected point gains for both sides
  - **Forecasts total team performance** over time
  - **And most importantly, the system provides a quantitative basis for decision-making**

CONCLUSION

- Next Steps
  - Fully integrate the Deep Q-Network to improve on player trade suggestions, improving on the current rule-based logic
    - Including better, more comprehensive state encoding for the agent
  - Automate roster updates
  - Improve the UI
  - Work through some of the complications with the FPL Draft API

REFERENCES

[1] V. Rajesh, P. Arjun, K. R. Jagtap, S. C. M and J. Prakash, "Player Recommendation System for Fantasy Premier League using Machine Learning," *2022*

[2] A. Peddii and R. Jain, "Random Forest-Based Fantasy Football Team Selection," *2023*

[3] M. Bangdiwala, R. Choudhari, A. Hegde and A. Salunke, "Using ML Models to Predict Points in Fantasy Premier League," *2022*